

ASG CAD DRAFTING SOP - HOW TO USE THE MASTER TEMPLATE

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C. | AL SIRAJ GROUP | Aluminium, Glass & Facade Division
Applies to: every estimator / draftsman producing AutoCAD drawings

Doc No. ASG-SOP-CAD-001

SOP Rev 0 | Standard R1

Date 10.10.2026

1. EVERY NEW DRAWING - STEP BY STEP

- 1. NEW > ASG_Master_Template.dwt.** Never start from acad.dwt or an old project file.
- Save at once with the correct name: **drawing number + rev + short project name**, e.g. AS-2026-014-ALU-SD-001-R0_MarinaTower.dwg
- Draw in the **Model** tab, **full size 1:1 in millimetres**. A 2400 mm opening is drawn 2400 long.
- Pick the correct **layer** before drawing (table 2). Never draw on layer 0.
- Open the sheet tab you need: A4-LANDSCAPE, A4-PORTRAIT, A3-LANDSCAPE, A3-PORTRAIT, A1-LANDSCAPE. Delete the tabs you do not use.
- Unlock the viewport, set its scale from the list (1:1, 1:2, 1:5, 1:10, 1:20, 1:25, 1:50, 1:100), **lock** it again.
- Double-click inside the viewport and add notes, dimensions and leaders with the ASG styles (table 3). Type the **paper height** (2.5) - AutoCAD sizes it for the viewport scale.
- Double-click the title block and fill every field. Never explode or move it.
- Plot: the page setup is ready (DWG To PDF, A-size paper, 1:1, ASG_Monochrome.ctb). Click Plot, then **open the PDF and check it**.

2. WHAT GOES ON WHICH LAYER

What you draw	Layer
Frame / panel / opening outline (elevation, plan)	ASG-OUTLINE-PRIMARY
Edges beyond, minor lines	ASG-OUTLINE-SECONDARY
Aluminium profiles, beads	ASG-ALUMINIUM-PROFILE
Glass, mirror	ASG-GLASS
Mild steel / stainless steel	ASG-MILD-STEEL / ASG-STAINLESS-STEEL
Hardware / anchors, screws, brackets	ASG-HARDWARE / ASG-FIXING
Gaskets / silicone, backer rod	ASG-GASKET / ASG-SEALANT
Insulation, thermal break	ASG-INSULATION
ACP / panels, flashings	ASG-CLADDING
Outline of material cut in a section	ASG-SECTION-CUT
Hatch / solid fill in sections	ASG-HATCH / ASG-HATCH-FILL
Hidden edges / centrelines	ASG-HIDDEN / ASG-CENTERLINE
Section cut line / section & detail marks	ASG-CUTTING-PLANE / ASG-SYMBOL
Setting-out lines that must print	ASG-SETOUT
Architect / consultant background (xref)	ASG-REFERENCE
Helper lines - NOT printed	ASG-CONSTRUCTION
Notes / leaders / tables / levels / grids	ASG-TEXT-NOTE / ASG-MULTILEADER / ASG-TABLE / ASG-LEVEL / ASG-GRID
Revision cloud + delta	ASG-REVISION
Reminders for yourself - NOT printed	ASG-NOPLLOT

Colours on screen show the job of the line; on paper everything prints black (hatch, insulation and background print grey). Line thickness comes from the layer - never change colour, linetype or lineweight of an object (keep **ByLayer**).

3. WHICH STYLE TO USE

Item	Style	Use
Dimension	ASG-DIM-ANNO	Normal dims - whole mm (1200)
Dimension	ASG-DIM-DETAIL	Fabrication details - 0.0 (50.0)
Dimension	ASG-DIM-ANGULAR / -RADIUS / -DIAMETER	30.00 deg / R40 / Ø80
Dimension	ASG-DIM-FIXED	Only when dimensioning in the layout tab
Leader	ASG-ML-ANNO / -DETAIL / -REFERENCE	Normal / dense detail / 'see drawing ...'
Note text	ASG-TEXT-NOTE 2.5 (small: ASG-TEXT-SMALL 1.8)	Model tab
Titles	ASG-TEXT-TITLE 3.5 / ASG-TEXT-HEADING 5.0	Layout tab
Tables	ASG-TABLE-STANDARD, -MATERIAL, -GLASS, -HARDWARE, -REVISION, -REGISTER, -NOTES	Layout tab

4. GOLDEN RULES

- Only the template, only ASG layers and ASG styles. Never use the 'Standard' styles.
- Never type over a dimension value.** If the number is wrong, the drawing is wrong - fix the lines.
- Viewports always **locked**. Unlock only to change the scale.
- Title block: never explode, scale or move it.
- No new layers or styles on your own - ask the CAD in-charge.
- Consultant / architect drawings: attach as xref (Overlay, relative path) on ASG-REFERENCE.
- Profile codes, glass make-ups, hardware: only from approved supplier catalogues - never guess.

5. BEFORE YOU ISSUE - CHECKLIST

- ☐ All title block fields filled; drawing number in the company format; REV correct.
- ☐ Every changed area has a revision cloud + delta, and the REV / revision list is updated.
- ☐ Nothing on layer 0; no object with its own colour / linetype / lineweight.
- ☐ Viewport scales correct, locked, and written in the SCALE field.
- ☐ Text and dimensions readable; nothing overlapping; no overridden dimension values.
- ☐ Helper lines only on ASG-CONSTRUCTION / ASG-NOPLLOT (they do not print).
- ☐ **AUDIT** (Y) and **PURGE** (all) done, file saved.
- ☐ PDF opened and checked: right paper size, lines thick/thin as expected, nothing cut off.

6. PROBLEM - QUICK FIX

Problem	Fix
All lines print equally thin	ASG_Monochrome.ctb missing in Plot Styles folder - tell CAD in-charge (meanwhile use monochrome.ctb)
Text / arrows huge or tiny	Wrong style or scale: use ASG-*-ANNO styles and set the viewport scale first
Note missing in one viewport	Select it > right-click > Annotative Object Scale > Add that scale
Viewport drawing moved / zoomed	Viewport was unlocked: unlock, set scale again, lock
Helper lines appear on the PDF	Move them to ASG-CONSTRUCTION or ASG-NOPLLOT
Title block damaged	Copy a clean one from a new drawing made from the template

7. DRAWING NUMBER

AS-[YYYY]-[NNN]-[DIS]-[TYP]-[SSS] + REV R0, R1, R2 ...
DIS: ALU aluminium | GLZ glazing | MS mild steel | SS stainless steel | MISC other metal works

TYP: GA general arrangement | EL elevation | SD shop drawing | SEC section | DT detail

Status: PRELIMINARY, FOR APPROVAL, APPROVED, FOR CONSTRUCTION, FOR FABRICATION, AS-BUILT, SUPERSEDED

8. ONE-TIME SETUP (CAD IN-CHARGE ONLY)

- Copy **ASG_Monochrome.ctb** to the Plot Styles folder; open it: every colour = 'Use object lineweight'.
- OPEN ASG_Master_Template.dxf > AUDIT > APPLOAD ASG_Setup.lsp > type **ASG-SETUP**.
- Last line must read **0 FAIL** (log: ASG_QA_Log.txt). ASG-QA checks the **template** only - it is not a check for project drawings.
- SAVEAS .dwg, then SAVEAS .dwt; put DWT + CTB on the shared drive; set QNEW on every PC.

Full rules: **ASG_CAD_Standards.pdf** (Standard R1). Questions / changes: CAD in-charge _____

Prepared by: _____ Reviewed by: _____ Approved by: _____

Controlled document - print and keep at your workstation